

Physical activity based prediction of hypoglycemia and hyperglycemia in youth with type-1 diabetes

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Abstract

Objective: To predict post-exercise glucose conditions within two hours in pediatric patients with type 1 diabetes (T1D), considering the impact of daily activity and demographic variability, including age-adjusted BMI. **Methods:** We used demographic information, CGM-derived metrics, and physical activity data from 241 adolescents with T1D, obtained from the T1DEXIP dataset. Machine learning models, including Mixed Effects Random Forest (MERF) and Gaussian Process Boosted Trees (GPBoost), were applied to account for repeated measures and within-subject variability. A deep learning model (LSTM) was also developed using raw time-series inputs. Pre-exercise windows of 2, 4, and 6 hours were compared. **Results:** The 6-hour GPBoost model demonstrated the best overall performance, particularly in identifying hypoglycemia events (with a recall of about 60%), despite persistent class imbalance. LSTM models also showed improvement with longer input windows but struggled with hypoglycemia detection due to limitations in oversampling sequential data. **Conclusion:** Integrating pre-exercise physiological and contextual data can enhance prediction of post-exercise glycemic events. The findings support the use of extended time windows and interpretable tree-based models for clinical decision-making in pediatric T1D management.

Keywords

Continuous Glucose Monitor, Daily Activity, Glucose Event, Physical Activity, Predictive Modeling, Type-1 Diabetes

Introduction

Predicting blood glucose variability is a complex task influenced by various physiological and behavioral factors. Adolescents aged 12 to 18 years with type 1 diabetes (T1D) are particularly vulnerable to unstable glycemic patterns due to increased physical activity and hormonal changes associated with puberty. Clinically, hyperglycemia is typically defined as blood glucose levels exceeding 180 mg/dL for more than 15 minutes, while hypoglycemia refers to levels below 70 mg/dL for the same duration. Recurrent episodes of hyperglycemia or hypoglycemia during adolescence can lead to short-term complications such as dizziness or seizures, as well as long-term health risks including cardiovascular disease and cognitive impairment¹⁰. Given the clinical importance of these risks, a growing number of studies have sought to track blood glucose fluctuations in relation to physical activity under real-world conditions.

Several studies have actively investigated postexercise glycemic outcomes in the context of daily life. For instance, Prasanna et al. (2023)⁷ examined how pre-exercise glucose levels and adjusted area under the glucose curve (AUC) influence the likelihood of hypoglycemia across various types of physical activity. Bergford et al. (2024)² developed predictive models for hypoglycemia and hyperglycemia risk during and after physical activity using real-world data from adolescents with T1D. Additionally, Sherr et al. (2024)⁸ identified key factors associated with post-exercise glycemia and nocturnal hypoglycemia through a large-scale pediatric cohort. While these studies primarily focus on post-exercise glycemic outcomes, the present study extends the analytical

perspective by incorporating the role of routine physical activity prior to exercise. Specifically, we define three pre-exercise windows, 2 hours, 4 hours, and 6 hours, to evaluate which timeframe offers the most predictive value for post-exercise glycemic events.

This study aims to predict the risk of hyperglycemia and hypoglycemia within two hours after physical activity by utilizing continuous glucose monitoring (CGM) metrics, demographic information, and daily activity data from the 2-, 4-, and 6-hour pre-exercise periods, all collected during the preceding 24 hours. Through our analysis of the data we had, we found that 2 hours after physical activity is the time where a lot of glucose changes and events are seen and is the best time to monitor and take action. The CGM data were initially stored in long format to preserve the temporal structure, but were reshaped into wide format to enable the use of conventional machine learning algorithms. This transformation, however, introduced a methodological challenge: each subject could contribute multiple exercise events, violating the assumption of independent and identically distributed (i.i.d.) observations. To address this issue, we applied models that can account for repeated measures and within-subject correlation. Specifically, we

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used Mixed Effects Random Forest (MERF)⁵ and Gradient Boosted Decision Trees with Gaussian Process (GPBoost)⁹ to capture individual-level variability. Additionally, to model the temporal dynamics of glucose levels, we implemented a deep learning model based on Long Short-Term Memory (LSTM)⁶ model. To compare the reliability and modeling efficiency of different approaches, we applied both machine learning and deep learning techniques and evaluated their predictive performance.

Methods

Study Design and Cohort Description

This study was designed as a real-world observational study involving adolescents diagnosed with type 1 diabetes (T1D). A total of 241 participants aged 12 to 18 years were enrolled, including 50 individuals who had been newly diagnosed within the past year. Among the participants, 139 (approximately 52%) were male and 102 (approximately 42%) were female.

All participants were instructed to wear continuous glucose monitors (CGMs) for approximately 10 consecutive days. Insulin therapy was administered through either automated insulin delivery systems, insulin pumps, or multiple daily injections (MDI). Insulin dosage data were recorded via the Bant mobile application, while physical activity metrics, including type, intensity, and duration, were collected using Garmin wearable devices and self-reported entries through the app.

In total, 3,400 physical activity events were identified for analysis. Among these, approximately 96% were associated with a single glycemic response, normal, hypoglycemia, or hyperglycemia, within the 2-hour post-exercise window. The remaining 4% included complex cases involving multiple glycemic events, such as hypoglycemia followed by hyperglycemia, each lasting longer than 15 minutes.

To manage this complexity and ensure consistency, we defined the final label for each event based on the first glycemic condition observed within the 2-hour window. This decision simplifies classification and aligns with clinical practices where early interventions are guided by the initial glycemic response. As shown in Table 1, the label distribution reveals substantial class imbalance, with normal glycemia (n=1,567; 46.1%), hypoglycemia (n=285; 8.4%), and hyperglycemia (n=1,548; 46.5%).

Table 1. Distribution of the target variable.

Class	Label	Count (%)
Normal	0	1567 (46.1%)
Hypoglycemia	1	285 (8.4%)
Hyperglycemia	2	1548 (46.5%)

Data source

The data used in this study come from the T1-DEXIP Pediatric dataset, developed under the T1D Exercise Initiative funded by the Leona M. and Harry B. Helmsley Charitable Trust. This dataset was specifically designed to address the challenges of glycemic control during physical activity in youth with T1D and was collected

under free-living, real-world conditions. In addition to CGM-derived glucose values, the dataset includes a wide range of physiological and behavioral variables such as insulin therapy details, physical activity types and intensities, exercise duration, and nutritional intake. These features enable a comprehensive assessment of how physical activity influences glucose dynamics in youth with T1D and serve as a foundation for predictive modeling of glycemic events.

Data Preprocessing

To support traditional machine learning models such as MERF and GPBoost, the dataset was transformed from long format (5-minute interval rows) to wide format. In the long format, each row represented a single timestamped measurement, which preserved temporal granularity but was not compatible with tabular model input. To restructure the data, we aggregated time-series variables into 15-minute intervals over 6 hours prior to each exercise session, and then flattened them into a single-row feature vector.

Specifically, physiological variables such as glucose and heart rate were averaged, while insulin dosage, distance, steps, and calories were summed within each interval. This transformation resulted in 24 columns per variable (i.e., 4 columns/hour $\times$ 6 hours), generating a total of 144 time-series-based features. These were combined with demographic and contextual features to form the final input. Categorical variables (e.g., sex, insulin method) were one-hot encoded, and continuous variables were normalized. Repeated measures were preserved by including subject identifiers, allowing the modeling process to account for within-subject correlations.

For the time series-based modeling approach, we prepared two input streams for the LSTM model. The first stream consisted of sequential physiological variables, including CGM values, heart rate, and insulin dosage, collected prior to each exercise event. These variables were preserved in their original temporal order to capture time-dependent patterns relevant to glycemic variability. The second input stream contained contextual information for each exercise session, such as duration, intensity, and total calories burned. This contextual input was included only for exercise sessions that lasted more than five minutes.

Each time series was truncated or zero-padded to ensure uniform sequence length across samples. Input features were normalized within subjects to reduce inter-individual variability. The model input was constructed using sliding windows aligned to the start of each exercise event, allowing the model to learn from a fixed time window preceding physical activity.

Imputation of Missing Values

To address missing values, linear interpolation was applied to CGM data with gaps shorter than 4 hours, while longer gaps were excluded. For heart rate data, we tested both linear interpolation and a time-aligned mean across multiple days, and selected the latter due to its higher consistency and accuracy for modeling.

Feature Engineering

To enhance the predictive power of our models, we computed a set of CGM metrics that are widely accepted by

international clinical consensus for use in diabetes research and management¹.

The engineered CGM features include:

- **Time in Range (TIR):** Percentage of time glucose levels are within 70–180 mg/dL.
- **Time Below Range (TBR):** Percentage of time glucose levels are below 70 mg/dL, including <54 mg/dL (severe hypoglycemia).
- **Time Above Range (TAR):** Percentage of time glucose levels are above 180 mg/dL, including >250 mg/dL (severe hyperglycemia).
- **Lability Index:** Sum of absolute glucose changes between consecutive readings, indicating instability in glucose levels.
- **Rate of Glucose Change:** Average of absolute glucose changes per 5-minute interval.
- **Number of Excursions:** Count of glucose readings outside the safe range (<40 or >300 mg/dL).
- **Glucose Management Indicator (GMI):** An estimated HbA1c derived from the mean glucose level.
- **Glycemic Risk Index (GRI):** Composite risk score capturing deviations across the entire glucose range.
- **HBGI & LBGI:** Log-transformed risk indicators for hyperglycemia and hypoglycemia, respectively.

The formulas and thresholds used to calculate these metrics were adapted from clinical guidelines and literature and are summarized in a supplemental table. These metrics were calculated using two strategies:

1. *Previous-day CGM metrics*, summarizing the day before exercise.
2. *Pre-exercise window metrics*, summarizing the 24 hours leading up to each physical activity session.

These features served as critical inputs for the MERF, GPBoost, and LSTM models. Their inclusion helped capture temporal patterns in glycemic control and enhanced interpretability in assessing glucose variability and risk.

In this study, Body Mass Index (BMI) was manually calculated using sex, age, height, and weight. For youth participants, BMI was adjusted for age using the CDC Extended BMI-for-Age Growth Charts, considering the substantial age-related variability in growth patterns³. The BMI z-score was subsequently derived based on the calculated BMI, using the CDC data files that correspond to the extended growth charts⁴.

Model Development

Our analysis pipeline consists of three major stages: data processing, model development, and model evaluation. These stages reflect the logical flow of transforming raw data into predictive insight. The overall process is illustrated in Figure 1, which outlines the key steps in this study.

In the data processing stage, we performed cleaning and formatting of raw data collected from CGM sensors, heart rate monitors, and physical activity trackers. This included handling missing values, computing CGM-derived metrics, and restructuring the data into formats suitable for both conventional machine learning and time-series deep learning models. Next, in the model development stage,

we applied a hybrid approach combining machine learning and deep learning techniques. We trained models to predict post-exercise glycemic events using features derived from demographic, physiological, and pre-exercise activity data.

Finally, model evaluation involved comparing the predictive performance across modeling strategies using metrics such as F1-score, AUROC, and sensitivity. This comparative evaluation enabled us to identify which time window (2hr, 4hr, or 6hr before exercise) and which modeling technique offered the best trade-off between interpretability and accuracy in forecasting glycemic outcomes.

Our analytical pipeline is summarized in Figure 1. This study addresses a multiclass classification problem in which the target is the glycemic state within two hours after physical activity. The outcome variable (label) was defined as the first glycemic state observed within the two-hour window following the end of exercise, aligning with the clinical relevance of early intervention. As noted earlier, the class distribution revealed a notable imbalance, particularly with hypoglycemia cases being much fewer in number.

The input features can be categorized into four groups. First, CGM-derived metrics over the previous full day (24 hours), as described in the **Feature Engineering** Section, which summarize glycemic distribution, variability, and risk. Second, demographic information including age, sex, height, weight, and BMI z-score. Third, daily activity metrics from the 2-, 4-, and 6-hour windows prior to each exercise session, which capture physiological responses such as heart rate, calorie expenditure, activity duration, and intensity, collected via Garmin devices and the Bant application. Fourth, raw CGM and heart rate time series data used exclusively for the LSTM model.

These features were formatted as continuous sequences for the LSTM model, while aggregated summaries in wide-format were used for the GPBoost and MERF models.

Model Architecture and Training Strategy

Since both MERF and GPBoost are designed for binary classification tasks under the assumption of repeated measures, they cannot be directly applied to multiclass targets. Therefore, we adopted the One-vs-Rest (OvR) strategy to decompose the three-class glycemic prediction problem into multiple binary sub-problems. This allowed us to build separate classifiers for hypoglycemia, normal, and hyperglycemia outcomes, facilitating model training while maintaining interpretability.

We employed three models in total: two tree-based machine learning models capable of handling repeated measures (MERF and GPBoost), and one deep learning model based on Long Short-Term Memory (LSTM). For MERF and GPBoost, lagged features were constructed at 15-minute intervals and transformed into wide-format input. Subject groups were defined based on combinations of sex and BMI z-score to account for within-subject correlation.

The LSTM model accepted raw sequential inputs. A bidirectional LSTM layer was followed by a unidirectional LSTM layer to effectively capture both past and future dependencies. The output of the LSTM layers was concatenated with summary-level activity features and passed through fully connected layers. Dropout (rate = 0.3) was applied to each dense layer to prevent overfitting.

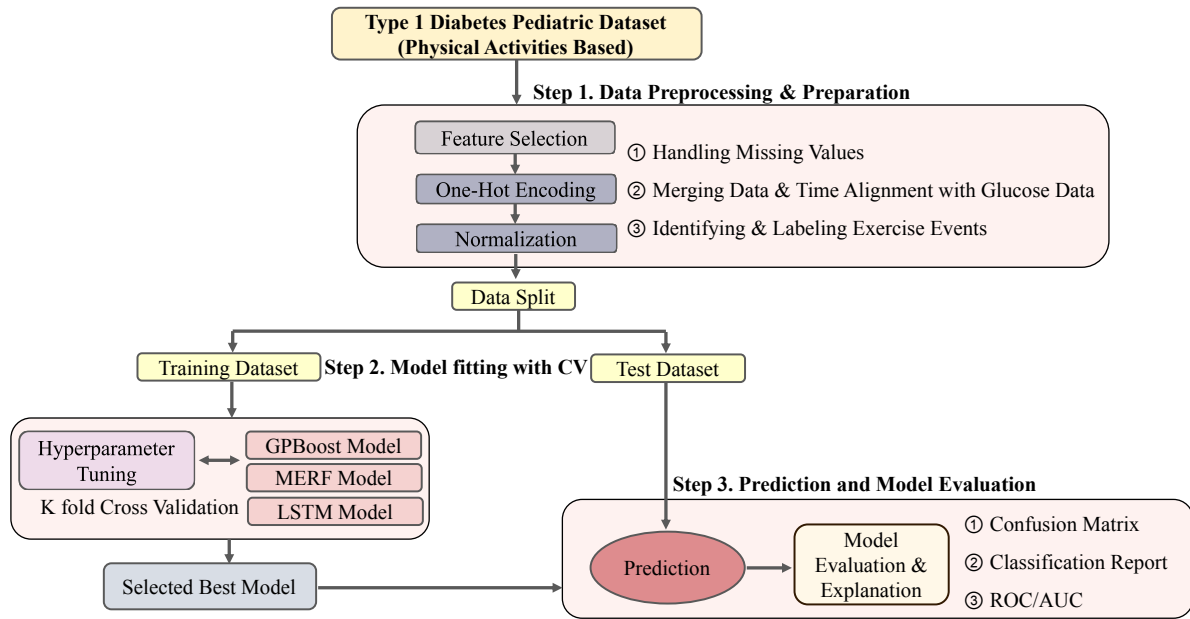


Figure 1. Flowchart for modeling.

by randomly dropping neurons during training, thereby promoting model generalization.

The detailed hyperparameter settings for each model are summarized in Table 2. All model development and training procedures were conducted using Python 3.6.8.

Table 2. Model hyperparameters and training settings.

Model	Hyperparameters / Settings
MERF	<code>n_estimators = 100</code>
GPBoost	<code>objective = binary</code> <code>learning_rate = 0.1</code> <code>max_depth = 6</code> <code>verbose = 0</code>
LSTM	<code>Dropout rate = 0.3</code> <code>loss:</code> <code>sparse_categorical_crossentropy</code> <code>optimizer: AdamW (lr=0.001)</code> <code>metric: accuracy</code>

Validation Strategy and Class Imbalance Handling

All models were trained using 5-fold Stratified Group K-Fold cross-validation. This strategy ensured that the same participant ID did not appear in both the training and test sets simultaneously, and that key variables such as BMI, sex, and glycemic state were evenly distributed across folds.

To address class imbalance, we applied different resampling strategies based on the model type and the classification target. For the MERF and GPBoost models, which followed a One-vs-Rest (OvR) approach, undersampling was performed only for the hypoglycemia vs. rest classification task, where the majority class (non-hypoglycemia cases) was reduced while maintaining the original class distribution. For the LSTM, oversampling was applied to the hypoglycemia class across all folds. This hybrid resampling approach helped improve recall and contributed to more balanced predictive performance across glycemic categories.

Evaluation Metrics

To evaluate the performance of our classification models, we employed several standard metrics commonly used in multiclass classification tasks. These metrics provide a comprehensive understanding of model behavior across different glycemic states:

Confusion Matrix: A tabular representation that summarizes the number of true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN) for each class. It helps identify specific classes that the model tends to misclassify.

Precision: Defined as the proportion of correctly predicted positive cases among all predicted positives. It quantifies how reliable the model is when predicting a particular class.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall (Sensitivity): The proportion of correctly predicted positive cases among all actual positives. This metric reflects the model's ability to detect true occurrences of a class.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-Score: The harmonic mean of precision and recall. It provides a balanced assessment, especially when the class distribution is imbalanced.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

For the machine learning models, we employed the OvR strategy, which involved training separate binary classifiers for each glycemic class: hypoglycemia, normal, and hyperglycemia. Accordingly, evaluation metrics such as precision, recall, and F1-score were computed independently for each class. In contrast, the LSTM model was trained as a multiclass classifier that simultaneously predicted all three glycemic states. Therefore, the evaluation metrics for the LSTM model were calculated in a unified manner.

We used macro-averaging to summarize overall performance. This approach computes the metric for each class independently and then takes the unweighted average across classes. Macro-averaging ensures that minority classes (such as hypoglycemia) are equally represented and not overshadowed by majority class performance.

Results

Table 3 summarizes the performance of the two models across different time windows. For the normal vs. rest and hyper vs. rest tasks, the models generally achieved an accuracy of approximately 60%. However, for the hypo vs. rest task, both models consistently exhibited a markedly low precision, indicating difficulty in accurately identifying hypoglycemia events, a pattern likely attributable to the substantial class imbalance. For the model selection, we tried to focus on a higher recall of hypoglycemia events, while maintaining strong overall accuracy due to the severe clinical consequences of hypoglycemia. Among all models, the 6-hour GPBoost model demonstrated the best performance, with a precision of approximately 13% and a recall of around 60%, outperforming other configurations. Accordingly, the GPBoost model trained on the 6-hour time window was selected as the final model.

The LSTM model trained for three-class classification exhibited varying performance depending on the sampling strategy, CGM metric type, and input time window. As shown in Table 4, oversampling consistently enhanced model performance across all configurations. The 6-hour input window using previous-day CGM metrics with oversampling yielded the highest accuracy of 0.49. The model achieved better performance with longer pre-exercise input windows, suggesting that additional historical glucose information contributes to improved classification accuracy. These results highlight the importance of selecting an appropriate sampling strategy and input window for enhancing glucose event prediction.

The Figures 2 and 3 present the averaged Receiver Operating Characteristic (ROC) curves and the confusion matrix, respectively, for the GPBoost model trained using a 6-hour input window. Each ROC curve corresponds to a binary classification task under the OvR framework, and the confusion matrix results are aggregated across all cross-validation folds. The GPBoost model's performance varied across the three binary classification tasks under the OvR framework. The Normal vs. Rest task yielded an AUC of 0.68, indicating a moderate ability to discriminate normal glycemic responses from abnormal ones. In contrast, the Hypoglycemia vs. Rest task produced the lowest AUC of 0.64, reflecting the inherent challenge in accurately identifying hypoglycemia events, likely due to class imbalance and the subtle temporal dynamics of glucose fluctuations. On the other hand, the Hyperglycemia vs. Rest task achieved the highest AUC of 0.73, suggesting that the model was most effective in detecting hyperglycemic responses. Despite correctly identifying 178 hypoglycemic cases, the confusion matrix revealed that the model misclassified 1,205 non-hypoglycemic instances as hypoglycemia, which substantially lowered the model's precision for this class. Moreover, 107 actual hypoglycemia

cases were not detected (i.e., false negatives), pointing to the model's limited sensitivity in recognizing this clinically important minority class.

The observed improvements over this baseline demonstrate the predictive value of pre-exercise data. These results underscore the persistent challenge of modeling hypoglycemia in imbalanced datasets and the need for refined strategies to improve recall without sacrificing precision.

Discussion and Conclusion

The primary objective of this study was to identify the most effective pre-exercise time window for predicting post-exercise glycemic events, with a particular focus on hypoglycemia, which poses severe clinical risks. To achieve this, we compared multiple modeling strategies, including the MERF and GPBoost models and a LSTM model, across 2-, 4-, and 6-hour pre-exercise windows.

As shown in Table 3, the tree-based model GPBoost exhibited relatively consistent predictive performance across time windows for the Normal vs. Rest and Hyperglycemia vs. Rest tasks. This suggests that for these outcomes, the length of the input window may have limited influence. However, in the Hypoglycemia vs. Rest classification, model performance varied significantly across time windows and modeling strategies. The 6-hour GPBoost model showed the best overall performance in this task, highlighting the importance of selecting an appropriate time window to improve recall in hypoglycemia prediction. Despite implementing undersampling techniques to address class imbalance, the model still struggled with low precision and recall for the hypoglycemia class. This suggests that the resampling strategy may not have been sufficient to effectively capture the minority class in the presence of high-dimensional and complex input features.

As presented in Table 4, the LSTM model also showed improved performance when trained with a longer pre-exercise time window. However, unlike GPBoost, the LSTM model demonstrated a noticeably lower recall for hypoglycemia cases, despite applying oversampling techniques. This indicates that many at-risk events were not correctly identified, which limits the model's clinical applicability. One potential explanation lies in the nature of time-series data. Unlike tabular input used in tree-based models, time-series models like LSTM must learn temporal dependencies and multivariate correlations. Oversampling strategies, which are often developed for static tabular data, may have generated synthetic sequences that failed to retain the intrinsic structure of real glucose trajectories. Consequently, the ability to detect rare glycemic patterns such as hypoglycemia was limited.

In conclusion, the results emphasize the critical role of input window selection and sampling strategy in building effective predictive models for glycemic events. While tree-based models like GPBoost offer robust performance and better interpretability for classifying hypoglycemia using structured inputs, time-series deep learning models may require more sophisticated augmentation techniques to fully leverage their potential. Future work should focus on developing advanced resampling methods tailored to multivariate time-series inputs, as well as incorporating

Table 3. Performance metrics for GPBoost and MERF models.

Model	Time Window	Class vs. Rest	Precision	Recall	F1-score	Accuracy
GPBoost	2-hour	Normal	0.60	0.57	0.58	0.62
		Hypoglycemia	0.12	0.58	0.20	0.63
		Hyperglycemia	0.66	0.60	0.63	0.67
	4-hour	Normal	0.61	0.59	0.59	0.63
		Hypoglycemia	0.11	0.53	0.19	0.63
		Hyperglycemia	0.66	0.60	0.62	0.67
	6-hour	Normal	0.61	0.59	0.60	0.63
		Hypoglycemia	0.13	0.60	0.21	0.63
		Hyperglycemia	0.66	0.59	0.62	0.67
MERF	2-hour	Normal	0.60	0.62	0.61	0.63
		Hypoglycemia	0.12	0.50	0.19	0.64
		Hyperglycemia	0.66	0.61	0.63	0.68
	4-hour	Normal	0.60	0.61	0.60	0.63
		Hypoglycemia	0.12	0.52	0.19	0.64
		Hyperglycemia	0.65	0.60	0.63	0.67
	6-hour	Normal	0.60	0.61	0.60	0.63
		Hypoglycemia	0.12	0.54	0.20	0.64
		Hyperglycemia	0.65	0.60	0.62	0.67

Table 4. Performance metrics for the LSTM model.

Time Window	Target	Precision	Recall	F1-Score	Accuracy
2-hour	Normal	0.45	0.64	0.53	0.44
	Hypoglycemia	0.56	0.07	0.12	
	Hyperglycemia	0.43	0.53	0.47	
4-hour	Normal	0.49	0.57	0.52	0.44
	Hypoglycemia	0.50	0.06	0.08	
	Hyperglycemia	0.39	0.62	0.48	
6-hour	Normal	0.52	0.64	0.57	0.49
	Hypoglycemia	0.56	0.20	0.30	
	Hyperglycemia	0.44	0.55	0.49	

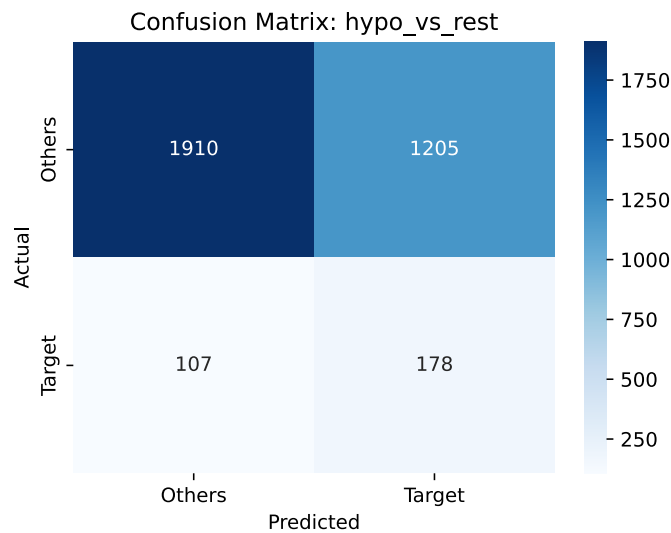


Figure 2. Confusion Matrix: Hypo vs. Rest (6-hour Window).

contextual or physiological signals that can help improve prediction sensitivity for rare but clinically important events.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

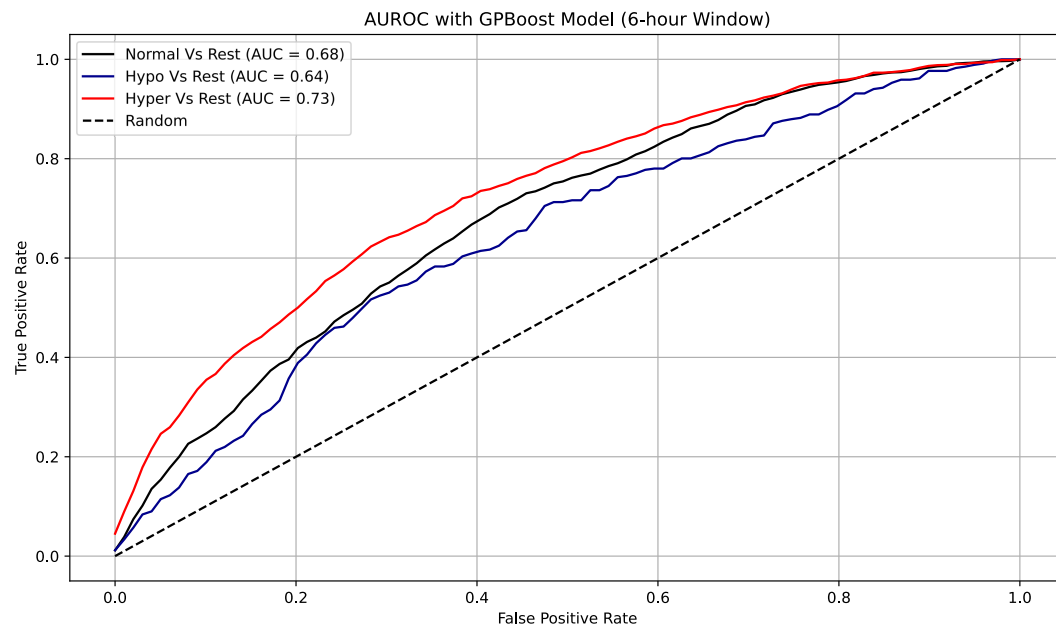


Figure 3. AUROC with GPBoost Model (6-hour Window).

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